

Optical-Axis Anisotropy on the A=1 Discrete Causal Lattice: Kinematic Dispersion and the Gauge-Sector Birefringence Cancellation*

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Data availability. The LaTeX source for this paper is publicly available at github.com/JackDMenendez/dcl-paper-04-optical-axis-birefringence.

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*Version 0.1-DRAFT. Paper IV of the A=1 Discrete Causal Lattice series; takes up follow-on item #14 (*Balanced Equations and Birefringent Channels*) of Paper I's [notes/follow_on_implications.md](#) catalogue. This paper subsumes the former companion study of the electric induced-action block (Paper VIII), merged here since the shared optical axis is the natural unity of the two effects. Builds on *Geometry First* (Paper I, DOI: 10.5281/zenodo.20613677) and *Geometry Forces Physics* (Paper II, DOI: 10.5281/zenodo.20292158). Pre-release draft; DOI to be assigned at v1.0 deposit.

Abstract

The A=1 discrete causal lattice of Paper I [1] singles out an optical axis $\hat{\mathbf{n}}_* = (1, 1, -1)/\sqrt{3}$ —the one cube body-diagonal absent from its bipartite hop set—and this single geometric fact organizes two distinct anisotropies, which we work to closed form. The first is *kinematic*: the single-tick spinor transfer operator has a structure-factor magnitude that is flat along $\hat{\mathbf{n}}_*$ and curved across it, with effective-mass tensor $M_{\text{eff}} = \frac{4}{3}P_{\perp}$, so propagation along the axis and across it are inequivalent. This is a uniaxial *directional dispersion anisotropy*—an axial–transverse contrast between different propagation directions—not a fixed-direction polarization split; we are careful to establish what the transfer-operator eigenphase does and does not license, since the operator is non-unitary off-axis. The second is a *gauge*-sector effect, and it is here that the term birefringence properly applies. Integrating the lattice’s response to a background field gives two geometric plaquette forms—an electric ϵ (eigenvalues $\{1, 4, 4\}$, axis suppressed) and a magnetic μ^{-1} (eigenvalues $\{4, 4, 16\}$, axis enhanced), read off the running engine—bound by the exact identity $\mu^{-1} = \text{adj}(\epsilon)$ (a theorem for any hop vectors). That closure forces the photon dispersion to a doubled transverse root: the two polarizations share one dispersion and the vacuum *birefringence cancels* (confirmed numerically to 10^{-15}), leaving a factor- ~ 2 common-mode speed anisotropy whose observability we leave open. We are explicit about what remains conditional: the plaquette forms are geometric candidates for the response blocks, and their derivation as the full dynamical induced action is a stated open item, so the null-birefringence theorem is rigorous *given* the blocks. Every claim’s scope, evidence, and status is recorded in the audit table (Table 1.1).

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1 Introduction

A crystal is optically anisotropic when its lattice singles out an axis. The $A=1$ discrete causal lattice of Paper I [1] singles out an axis too, but the “crystal” is spacetime itself: its bipartite hop set—the three RGB vectors $\mathbf{V}_1 = (1, 1, 1)$, $\mathbf{V}_2 = (1, -1, -1)$, $\mathbf{V}_3 = (-1, 1, -1)$ and their negatives—uses three of the four cube body-diagonals and leaves the fourth, $\hat{\mathbf{n}}_* = (1, 1, -1)/\sqrt{3}$, out. That omission breaks the cubic point group O_h to the trigonal D_{3d} about $\hat{\mathbf{n}}_*$ and fixes an optical axis geometrically, with no new postulate. This paper works the resulting anisotropy to closed form in the two places it appears—the matter propagator and the vacuum’s gauge response—which share the one axis.

The two effects are physically different, and we name them differently. The *kinematic* anisotropy (Sec. 2) follows from Paper I’s single-tick spinor transfer operator: its entire wavevector dependence passes through one scalar structure factor whose magnitude is flat along $\hat{\mathbf{n}}_*$ and curved perpendicular to it. This is a uniaxial *directional dispersion anisotropy*—a contrast between propagation *directions*, maximal across the axis and vanishing along it—rather than a polarization split at a fixed direction, which is what “birefringence” denotes; we reserve that word for the gauge sector. Because the transfer operator is non-unitary away from the axis, we are careful to separate what its eigenphase magnitude establishes (an exact axial–transverse anisotropy of the structure factor) from the further physical reading of it as a propagating-mode group velocity, which we state as requiring the $A=1$ normalization rather than assuming it.

The *gauge* anisotropy (Sec. 3) is the vacuum’s response to a background electromagnetic field, and here the outcome is the opposite of a new signal. The lattice supplies two geometric plaquette quadratic forms—an electric permittivity ϵ from the on-site temporal coupling and a magnetic inverse permeability μ^{-1} from the spatial Peierls holonomy—and they obey the exact algebraic identity $\mu^{-1} = \text{adj}(\epsilon)$. This closure makes the two transverse photon polarizations share a single dispersion, so the gauge-sector vacuum *birefringence cancels*; what remains is a direction-dependent common-mode speed affecting both polarizations equally, whose observational status is open. The adjugate theorem and its doubled-root consequence are the paper’s central algebraic result. We are equally explicit about the load-bearing caveat: the two forms are the natural *geometric* plaquette candidates for the response blocks, and establishing them as the full *dynamical* induced action—the second-order effective action obtained by integrating out the matter—is a step we do not complete, so the cancellation theorem is rigorous conditional on the proposed blocks.

This gauge-sector material was developed first as a separate companion study; it is merged here because the shared optical axis is the natural unity of the two effects. The gauge

blocks are additionally read off the running lattice engine and the null verdict re-derived through a separate code path, which we describe as an independent code-path verification—a useful cross-check within the same framework, engine, and geometric objects—rather than an independent physical measurement.

1.1 Scope

Derived here and settled: the closed-form kinematic directional dispersion anisotropy (Sec. 2); the adjugate constitutive closure and the null gauge-sector birefringence *conditional on the geometric blocks* (Sec. 3); and the structural axis-sharing of the two effects. Explicitly deferred: the derivation of the gauge blocks as the actual dynamical induced action (a matter-integrated effective action; the single-object $\text{Tr} \ln T$ tensor extraction is a stated open item); the relative electric/magnetic normalization γ that fixes the absolute vacuum speed (a common coupling cancels and does not set it); and the observational status of the common-mode speed anisotropy (the single-domain-versus- O_h -restoration question). The physical reading of the kinematic eigenphase as a photon energy dispersion, and the identification of the massless spinor mode with a photon, are likewise flagged where they exceed what is proven. The audit table (Table 1.1) is the canonical record of which claims are **PASS**, **PART**, **STUB**, or **FAIL**.

Table 1: System audit: paper claims mapped to supporting derivations and experiments. Status semantics in the front-matter (PASS / PART / STUB / FAIL).

Claim	Mechanism	Continuum / formal limit	Evidence	Status
Kinematic directional dispersion anisotropy along $(1, 1, -1)$ (axial-transverse contrast)	Uniaxial axial-transverse anisotropy of the structure-factor magnitude: $M_{\text{eff}} = \frac{4}{3}(\mathbb{I} - \hat{\mathbf{n}}_* \hat{\mathbf{n}}_*^T) = \frac{4}{3}P_{\perp}$, so $ H_{\text{RGB}}(\mathbf{k}) ^2 = 1 - \frac{4}{3} \mathbf{k}_{\perp} ^2$, flat along $(1, 1, -1)$. Massive-sector group speed $\frac{2}{3}\sin\omega \mathbf{k} \sin\alpha$ – a contrast between propagation <i>directions</i> , not a fixed- $\mathbf{k}$ polarization split (hence <i>not</i> birefringence)	Isotropic under O_h averaging / $a \rightarrow 0$	<i>Analytic:</i> $ H_{\text{RGB}} ^2 = 1$ exactly along $(1, 1, -1)$ at all orders; exp_02_closed_form_split_check (identities $< 10^{-12}$, speed coeffs $< 10^{-5}$). <i>Observability:</i> exp_01_optical_axis_isotropy – isotropic pulse flattens perpendicular to $(1, 1, -1)$, survives exact $A=1$ renormalization. <i>Interpretation caveat (what PASS covers):</i> the derived object is the exact structure-factor anisotropy; $T(\mathbf{k})$ is non-unitary off-axis, so the eigenphase- $\rightarrow$ -group-velocity reading requires the $A=1$ normalization argument, and at $\omega = 0$ the surviving anisotropy is of the eigenvalue <i>magnitude</i> (not yet a proven photon energy dispersion); the massless-spinor- $\rightarrow$ -photon identification is not carried out here.	PASS
Gauge-sector null polarization split (birefringence cancels), <i>conditional on the geometric blocks</i>	With $\boldsymbol{\mu}^{-1} = \gamma \text{adj}(\boldsymbol{\epsilon})$ the photon dispersion is a <i>double</i> transverse root $\det(K\boldsymbol{\mu}^{-1}K + \omega^2\boldsymbol{\epsilon}) = \det(\boldsymbol{\epsilon})\omega^2(\omega^2 - k^T\boldsymbol{\epsilon}k)^2$ for every direction, so the split $\propto \varepsilon_a\mu_a^{-1} - \varepsilon_p\mu_p^{-1} = 0$	Isotropic-impedance product $\varepsilon_i(\mu^{-1})_i = \gamma \det \boldsymbol{\epsilon}$; rescaling-invariant, so immune to the deferred normalization; holds for any hop vectors	<i>Adjugate theorem</i> $Q_B = \text{adj}(\boldsymbol{\epsilon})$, $\boldsymbol{\epsilon} = MM^T$, proven for general vectors (electric_induced_action.py , symbolic). <i>Engine + numeric:</i> exp_03_geometric_blocks confirms the adjugate from the two engine blocks (5×10^{-15}); the Fresnel solver exp_03_dispersion_core gives $\max v_+ - v_- = 1.6 \times 10^{-15}$ over 2000 directions (detuned control 0.5); 2×10^4 - k symbolic-guard sweep $< 10^{-14}$. An <i>independent code-path verification</i> (same framework/engine), not a separate physical measurement.	PASS

(continued on next page)

(Table 1.1, continued from previous page)

Claim	Mechanism	Continuum / formal limit	Evidence	Status
Gauge blocks as the actual <i>dynamical</i> induced action (ϵ , μ^{-1})	The two forms are the natural <i>geometric</i> plaquette quadratic forms: $\epsilon = \sum_a \mathbf{V}_a \mathbf{V}_a^T$ (temporal holonomy $\Theta_a = \mathbf{V}_a \cdot \mathbf{E}$, eig $\{1, 4, 4\}$), $\mu^{-1} = Q_B$ (spatial holonomy, eig $\{4, 4, 16\}$)	The dynamical induced action is a matter-integrated (loop-level) effective action; the plaquette forms are bare geometric actions	<i>Vertex verified:</i> $\delta\phi = \omega + V$ read off <code>HopOperator.step</code> (<code>exp_04_electric_permittivity_extraction</code>); fidelity 4×10^{-19} , ω cancels, $a_t = 1$. <i>Gap (why PART):</i> this validates the field-matter <i>vertex</i> , not the quadratic effective action $\Gamma^{(2)}$; a general D_{3d} -invariant form has two free coefficients, so the 1:4 ratio is not symmetry-forced. Promotion needs $\epsilon_{ij} \propto \partial^2 \Gamma / \partial E_i \partial E_j$ ($\Gamma = -\log \det T$) or a measured 2nd-order response; the $\text{Tr} \ln T$ route is Ward-divergent on the flat axis (naive band sum $\rightarrow$ wrong object about $(1, -1, 0)$) and is a stated open follow-on.	PART
Vacuum speed isotropy / O_h restoration (common-mode, factor ~ 2 about $(1, 1, -1)$)	Both polarizations share $v^2(\mathbf{k}) = \mathbf{k}^T \epsilon \mathbf{k} \in [1, 4]$; the lattice has <i>one fixed</i> hop-set (axis = the omitted 4th cube body-diagonal), a single trigonal domain	O_h 4-domain average $\langle \epsilon \rangle = 3\mathbb{I}$, $\langle \mu^{-1} \rangle = 8\mathbb{I}$ (isotropic), but a <i>fictitious ensemble</i> the fixed lattice does not realize; note the <i>exact</i> adjugate fails averaging ($\text{adj}(3\mathbb{I}) = 9\mathbb{I} \neq 8\mathbb{I}$) – only the proportional closure survives	<code>exp_03_geometric_blocks</code> : engine blocks give common-mode $v \in [1.00, 2.00]$ (single-domain factor 2.00); the average restores isotropy (10^{-16}). A <i>speed</i> anisotropy, not birefringence, and distinct from the null split. Whether the fixed lattice or an averaged ensemble is physical is undecided.	PART
Multi-channel concordance (P9)	Kinematic and gauge sectors inherit the <i>same</i> optical axis $(1, 1, -1)$ from the common bipartite geometry (M_{eff} , ϵ , μ^{-1} special directions coincide)	N/A (cross-sector structural identity)	Axis-sharing is a <i>structural</i> identity (derived): all three tensors inherit $(1, 1, -1)$ from the same geometry. The kinematic anisotropy is the one frontier-testable channel; the gauge leading birefringence <i>cancels</i> , so the shared axis registers there as the direction of the common-mode speed anisotropy, not a polarization split. A genuine structural concordance; its strength as a multi- <i>observable</i> claim is limited because the gauge birefringence is null and the common-mode speed's observability is open.	PART

2 Kinematic Dispersion Anisotropy along $(1, 1, -1)$

A terminological caution frames this section. The effect below is a uniaxial *directional dispersion anisotropy*: it contrasts propagation along $\hat{\mathbf{n}}_*$ with propagation across it. It is

not birefringence in the optical sense—two polarization eigenmodes with different speeds at one fixed wavevector—and we reserve that word for the gauge sector (Sec. 3). The two branches $\lambda_{\pm}(\mathbf{k})$ below are the eigenvalue branches of the spinor transfer operator (the paired positive/negative-frequency spinor modes), not two optical polarizations. A second caution concerns interpretation: the transfer operator is *non-unitary* away from the axis (Sec. 2.2, $|\lambda_{\pm}|^2 < 1$), so identifying the eigenphase gradient with a physical group velocity is a reading that the $A=1$ normalization must license rather than one that follows automatically. What is rigorous throughout is the exact axial–transverse anisotropy of the structure-factor magnitude; the group-velocity and photon readings are flagged where they exceed it.

The kinematic channel inherits Paper I’s single-tick spinor propagator [1]

$$T(\mathbf{k}) = \begin{pmatrix} i \sin(\omega/2) & \cos(\omega/2) \tilde{H}_{\text{RGB}}(\mathbf{k}) \\ \cos(\omega/2) \tilde{H}_{\text{CMY}}(\mathbf{k}) & i \sin(\omega/2) \end{pmatrix}, \quad \tilde{H}_{\text{CMY}} = \overline{\tilde{H}_{\text{RGB}}}, \quad (1)$$

whose eigenvalues $\lambda_{\pm}(\mathbf{k}) = i \sin(\omega/2) \pm \cos(\omega/2) |\tilde{H}_{\text{RGB}}(\mathbf{k})|$ carry the entire $\mathbf{k}$ -dependence through the single scalar $|\tilde{H}_{\text{RGB}}|$. Here ω is the per-tick Zitterbewegung rate ($m = \sin(\omega/2)/a$), and the massless photon is the limit $\omega \rightarrow 0$. The RGB basis vectors are $\mathbf{V}_1 = (1, 1, 1)$, $\mathbf{V}_2 = (1, -1, -1)$, $\mathbf{V}_3 = (-1, 1, -1)$, with $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 = (1, 1, -1)$ fixing the optical axis $\hat{\mathbf{n}}_* = (1, 1, -1)/\sqrt{3}$.

2.1 Exact structure factor and the perpendicular projector

With $\tilde{H}_{\text{RGB}}(\mathbf{k}) = \frac{1}{3} \sum_{\mathbf{V} \in \text{RGB}} e^{i\mathbf{k} \cdot \mathbf{V}}$ and the difference vectors $\mathbf{V}_1 - \mathbf{V}_2 = (0, 2, 2)$, $\mathbf{V}_1 - \mathbf{V}_3 = (2, 0, 2)$, $\mathbf{V}_2 - \mathbf{V}_3 = (2, -2, 0)$, the modulus is closed-form to all orders:

$$|\tilde{H}_{\text{RGB}}(\mathbf{k})|^2 = \frac{1}{9} \left[3 + 2 \cos 2(k_y + k_z) + 2 \cos 2(k_x + k_z) + 2 \cos 2(k_x - k_y) \right]. \quad (2)$$

Along the optical axis $\mathbf{k} = t(1, 1, -1)$ every argument vanishes, so $|\tilde{H}_{\text{RGB}}|^2 = 1$ *exactly for all* t : the dispersion is rigorously flat along $\hat{\mathbf{n}}_*$. Expanding to second order recovers Paper I’s M_{eff} , which takes the transparent form

$$|\tilde{H}_{\text{RGB}}(\mathbf{k})|^2 = 1 - \mathbf{k}^T M_{\text{eff}} \mathbf{k} + O(k^4), \quad M_{\text{eff}} = \frac{4}{3} (\mathbb{I} - \hat{\mathbf{n}}_* \hat{\mathbf{n}}_*^T) = \frac{4}{3} P_{\perp}. \quad (3)$$

That is, M_{eff} is $\frac{4}{3}$ times the projector onto the plane perpendicular to the optical axis (eigenvalues $\{4/3, 4/3, 0\}$). The dispersion therefore depends *only* on the perpendicular wavevector $\mathbf{k}_{\perp} = P_{\perp} \mathbf{k}$:

$$|\tilde{H}_{\text{RGB}}(\mathbf{k})|^2 = 1 - \frac{4}{3} |\mathbf{k}_{\perp}|^2 + O(k^4). \quad (4)$$

2.2 The eigenphase dispersion and the speed split

The mode energy (phase advance per tick) is $\theta(\mathbf{k}) = \arg \lambda_+(\mathbf{k})$. From Eq. (1), $\text{Re } \lambda_+ = \cos(\omega/2)|\tilde{H}_{\text{RGB}}|$ and $\text{Im } \lambda_+ = \sin(\omega/2)$, giving the exact dispersion

$$\tan \theta(\mathbf{k}) = \frac{\tan(\omega/2)}{|\tilde{H}_{\text{RGB}}(\mathbf{k})|}. \quad (5)$$

At $\mathbf{k} = 0$ this returns $\theta = \omega/2$ (the rest-mass phase); off-axis $|\tilde{H}_{\text{RGB}}| < 1$ raises θ . Differentiating gives the group velocity in closed form,

$$\mathbf{v}_g(\mathbf{k}) = \nabla_{\mathbf{k}} \theta = - \frac{\frac{1}{2} \sin \omega \nabla |\tilde{H}_{\text{RGB}}|}{\cos^2(\omega/2) |\tilde{H}_{\text{RGB}}|^2 + \sin^2(\omega/2)}, \quad (6)$$

which, using Eq. (4), reduces at small $\mathbf{k}$ to

$$\mathbf{v}_g(\mathbf{k}) = \frac{2}{3} \sin \omega \mathbf{k}_{\perp} + O(k^3). \quad (7)$$

Proposition 2.1 (Axial–transverse group-velocity contrast). *For a wavevector at angle α to the optical axis ($|\mathbf{k}_{\perp}| = |\mathbf{k}| \sin \alpha$), the kinematic group speed (read from the eigenphase, subject to the interpretation caveat below) is, to leading order,*

$$|\mathbf{v}_g| = \frac{2}{3} \sin \omega |\mathbf{k}| \sin \alpha. \quad (8)$$

The transverse direction ($\mathbf{k} \perp \hat{\mathbf{n}}_$, $\alpha = 90^\circ$) carries the maximal group speed $\frac{2}{3} \sin \omega |\mathbf{k}|$; the axial direction ($\mathbf{k} \parallel \hat{\mathbf{n}}_*$, $\alpha = 0$) is frozen, $|\mathbf{v}_g| = 0$. The axial–transverse contrast is*

$$\Delta v \equiv v_{\perp} - v_{\parallel} = \frac{2}{3} \sin \omega |\mathbf{k}|, \quad (9)$$

with $\hat{\mathbf{n}}_ = (1, 1, -1)/\sqrt{3}$ the frozen (axial) direction. This is a contrast between two propagation directions, not a polarization split at fixed $\mathbf{k}$.*

Remark 2.2 (Non-unitarity and what the eigenphase licenses). Away from the axis the transfer operator is not unitary: $|\lambda_{\pm}|^2 = \sin^2(\omega/2) + \cos^2(\omega/2) |\tilde{H}_{\text{RGB}}|^2 < 1$, so each eigenvalue carries amplitude renormalization as well as phase. Identifying $\nabla_{\mathbf{k}} \theta$ with a physical group velocity is therefore not automatic: it requires deriving wave-packet motion for this evolution including the momentum dependence of $|\lambda|$, or showing that the per-tick $A=1$ renormalization—which rescales $|\lambda|$ to unity at each step—converts $T(\mathbf{k})$ into a unitary physical propagator whose eigenphase is the energy. Pending that argument, the conservative statement of Eqs. (4)–(9) is that the structure-factor magnitude $|\tilde{H}_{\text{RGB}}(\mathbf{k})|$ has an exact

axial–transverse anisotropy; that it is an anisotropy of physical *transport* is corroborated observationally by `exp_01` (the isotropic pulse flattens perpendicular to $\hat{\mathbf{n}}_*$ under the full renormalized dynamics), which is the sense in which the reading is used here.

Equations (2), (5), (6) and Proposition 2.1 have been checked against independent direct computation in `exp_02_closed_form_split_check`: the structure-factor and dispersion identities to floating-point precision ($< 10^{-12}$, versus the triple sum and versus $\arg \lambda_+$ respectively), and the group velocity and principal speed coefficients $\frac{2}{3} \sin \omega$ and 0 to finite-difference truncation ($< 10^{-5}$). Together with `exp_01` these checks discharge the kinematic audit row to **PASS** (Table 1.1). The observability of the underlying anisotropy in renormalized amplitudes was established separately in `exp_01` (Table 1.1; `notes/optical_axis_renormalization_test.md`).

Remark 2.3 (The massless limit: what survives, and what it is not yet). The group-velocity contrast (9) scales as $\sin \omega$ and so vanishes as $\omega \rightarrow 0$. The limit is in fact subtle and must be stated carefully. At $\omega = 0$ the eigenvalues are $\lambda_{\pm} = \pm |\tilde{H}_{\text{RGB}}(\mathbf{k})|$, purely real, so the eigen*phase* is momentum-independent and the eigenphase group velocity is zero for *every* direction. What survives is the axial–transverse anisotropy of the eigenvalue *magnitude* $|\tilde{H}_{\text{RGB}}|$, Eq. (4). That magnitude curvature is, as it stands, an attenuation or filtering curvature of the transfer operator; turning it into a photon *energy* dispersion requires the additional physical argument of Remark 2.2 (how the $A=1$ normalization builds a unitary propagator whose eigenphase is the energy). Until that argument is supplied, the conservative massless claim is the exact axial–transverse anisotropy of the structure factor, not a demonstrated photon dispersion-curvature split. Relatedly, the massless limit of a spinor propagator is in the first instance a massless *spinor* mode; identifying the two-component mode with a photon—its helicity content, gauge character, and representation—is a separate step this paper does not carry out. The robust, framework-internal content of this section is the closed-form uniaxial anisotropy of $|\tilde{H}_{\text{RGB}}|$ about $\hat{\mathbf{n}}_*$ and its survival under $A=1$ renormalization (`exp_01`); the massive-sector transport reading (9) is its clearest observable instance.

Remark 2.4 (The split adds no propagating modes). The ordinary and extraordinary modes derived above are the two *direction-dependent phases* of the bipartite propagator’s existing eigenvalue branches $\lambda_{\pm}(\mathbf{k}) = i \sin(\omega/2) \pm \cos(\omega/2) |\tilde{H}_{\text{RGB}}(\mathbf{k})|$. The entire birefringence enters through the single scalar $|\tilde{H}_{\text{RGB}}(\mathbf{k})|$, whose low- $\mathbf{k}$ expansion $1 - \frac{4}{3} |\mathbf{k}_{\perp}|^2$ is flat along $\hat{\mathbf{n}}_*$ and curved (coefficient $4/3$) in the perpendicular plane. The closed-form split is thus a modulation of the *same* two branches—the (ψ_R, ψ_L) pair of the continuum Dirac spinor—and introduces no additional branch: the mode count is two, before and after.

This is logically independent of the lattice’s fermion-doubling content. Doublers are a Brillouin-zone-corner phenomenon, whereas the ordinary/extraordinary split is the second-

order term of an expansion about the physical mode at $\mathbf{k} = 0$; the two occupy disjoint regions of $\mathbf{k}$ -space. Paper I deliberately defers the doubler accounting for the bipartite T_{diamond} Dirac operator (Paper I §6.4, [1]) to a separate one-loop lattice-perturbation-theory calculation, and nothing in the present derivation depends on, alters, or settles that count. This $\mathbf{k}$ -space disjointness argument is inherited from Paper I’s framework and is not a new result of the present paper; in particular, the optical-axis birefringence does *not* imply a larger particle zoo.

Remark 2.5 (Vacuum birefringence and wave-packet spreading restate this channel). Two real-space signatures sometimes proposed as independent tests are, on this lattice, the *same* kinematic anisotropy read through different observables and add no new claim. First, a polarization/direction-dependent propagation speed—“vacuum birefringence”—is precisely the ordinary/extraordinary split of Proposition 2.1 and Eq. (4); it is this section’s subject, not a separate prediction. Second, the anisotropic *spreading* of an initially isotropic packet is the real-space dual of the group-velocity dispersion (6)—the same $\nabla_{\mathbf{k}}\theta$ that Eq. (9) names—and is exactly the observable that `exp_01` already measures (the pulse flattens perpendicular to $\hat{\mathbf{n}}_*$). The isotropic-spread rate itself (d^2E/dp^2 , direction-averaged) is the province of the Lorentz-violating dispersion program, whose real-space demonstrator lives there rather than here; only its $(1, 1, -1)$ -aligned *anisotropy* is this paper’s result. Any claim that the lattice packet spreads “faster than continuum QFT” must be stated as the residual that survives the continuum limit $a \rightarrow 0$: a free packet already spreads, and a lattice excess that vanishes as $a \rightarrow 0$ is numerical dispersion, not physics. What survives $a \rightarrow 0$ here is the *direction dependence* fixed by $\hat{\mathbf{n}}_*$, not an isotropic spread-rate enhancement.

3 Gauge-Sector Response and the Birefringence Cancellation along $(1, 1, -1)$

Where the kinematic channel (Sec. 2) lives in the matter propagator, the gauge channel lives in the vacuum’s response to a background electromagnetic field. It is here, and only here, that the paper speaks of *birefringence* in its literal optical sense: two transverse polarizations of one propagation vector, and whether their speeds split. The lattice couples the two static fields on different footings. A magnetic field enters as a *spatial* link phase—the hop along a lattice direction $\mathbf{V}_a$ picks up the Peierls phase $\exp(i\mathbf{A}\cdot\mathbf{V}_a)$, so a closed spatial plaquette carries a gauge-invariant Wilson holonomy. An electric field enters instead as an *on-site* phase advance $\delta\phi(\mathbf{x}) = \omega + V(\mathbf{x})$, the temporal potential A_0 added to the rest phase, with no spatial link. From these two couplings we build two quadratic field forms, show they are

bound by an exact algebraic identity, and read off its consequence for polarization.

A caveat frames everything below. The two forms are the natural *geometric* (plaquette) quadratic forms the hop set supplies; we call them a permittivity ϵ and an inverse permeability μ^{-1} because they play those roles in the constitutive relation, but establishing them as the *dynamical* induced action—the second-order response obtained by integrating the matter tokens out—is a separate step we do not complete here (it is the stated open item of Sec. 3.6). What is rigorous is the algebra of the two geometric forms and the polarization verdict it forces; that verdict is prefactor-independent, so it survives whatever fixes the blocks’ overall normalization.

3.1 The magnetic form (anchor)

A spatial plaquette spanned by two hop directions $\mathbf{V}_a, \mathbf{V}_b$ has, for a uniform field, holonomy $(\mathbf{V}_a \times \mathbf{V}_b) \cdot \mathbf{B}$; summing the squared holonomy over the three independent bipartite plaquettes gives the $O(B^2)$ form density $_B = \sum_{a < b} ((\mathbf{V}_a \times \mathbf{V}_b) \cdot \mathbf{B})^2 = \mathbf{B}^\top Q_B \mathbf{B}$, with

$$Q_B = \begin{bmatrix} 8 & 4 & -4 \\ 4 & 8 & -4 \\ -4 & -4 & 8 \end{bmatrix} \quad (10)$$

whose eigenvalues are $\{4, 4, 16\}$: the optical axis $(1, 1, -1)$ is *enhanced* (eigenvalue 16), the perpendicular plane doubly degenerate at 4. This is the Paper I Appendix B tensor [1], reproduced from the engine’s Peierls links by `dcl-core`’s `exp_04` [2] and again here (`exp_03_geometric_blocks`) to machine precision; it is the exact consistency anchor throughout (Table 1.1).

3.2 The electric form

The on-site advance $\delta\phi = \omega + V(\mathbf{x})$ is itself a *temporal* link—a site joined to itself at the next tick—so a temporal plaquette spanned by one hop direction $\mathbf{V}_a$ and the tick direction closes on the lattice. For a uniform static field $V(\mathbf{x}) = -\mathbf{E} \cdot \mathbf{x}$ its holonomy is the on-site phase difference along the hop,

$$\Theta_a(\mathbf{x}) = \delta\phi(\mathbf{x}) - \delta\phi(\mathbf{x} + \mathbf{V}_a) = V(\mathbf{x}) - V(\mathbf{x} + \mathbf{V}_a) = (\mathbf{V}_a \cdot \mathbf{E}) a_t, \quad (11)$$

the rest phase cancelling around the loop (a_t is the tick’s temporal extent). Where the magnetic loop needs two spatial directions and a cross product, the electric loop needs one. Summing the squared holonomy over the three hop directions gives density $_E = \sum_a \Theta_a^2 =$

$a_t^2 \mathbf{E}^\top \boldsymbol{\varepsilon} \mathbf{E}$, with

$$\boldsymbol{\varepsilon} = P = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix} \quad (12)$$

eigenvalues $\{1, 4, 4\}$: the same axis is now *suppressed* (eigenvalue 1), the perpendicular plane again at 4. The electric form is the mirror of the magnetic one—axis suppressed where the magnetic axis is enhanced—and its sense matches Paper IV’s static screen `exp_03a`.

This form is read off the engine’s actual tick evolution, not merely posited. Seeding a uniform real probe state and taking one step of the hop operator, the on-site term $i \sin(\delta\phi/2) \psi_R$ is purely imaginary while the kinetic hop is purely real, so $\text{Im}(\psi_R^{\text{new}})/\psi_R = \sin(\delta\phi/2)$ recovers $\delta\phi = \omega + V(\mathbf{x})$ from the step output (to 10^{-19}); $\boldsymbol{\varepsilon}$ is extracted from that recovered phase (`exp_04_electric_permittivity_extraction`), with the tick weight $a_t = 1$ measured, not assumed.

Remark 3.1 (What the read-off establishes, and what it does not). Recovering $\delta\phi = \omega + V$ from one `HopOperator.step` validates the field–matter *vertex* (the coupling in the tick rule), not the quadratic effective action $\Gamma^{(2)}[\mathbf{E}]$. A general D_{3d} -invariant quadratic form has *two* independent coefficients (an axial and a transverse one), so the 1:4 eigenvalue ratio is a property of the equal-weight single-hop plaquette sum, *not* something D_{3d} symmetry forces; cross-plaquette terms could reweight it. To promote $\boldsymbol{\varepsilon}$ from “geometric candidate for the electric response block” to “the induced permittivity” one needs the effective-action Hessian $\varepsilon_{ij} \propto \partial^2 \Gamma[\mathbf{A}]/\partial E_i \partial E_j|_{\mathbf{A}=0}$ with $\Gamma[\mathbf{A}] = -\log \det T[\mathbf{A}]$, or a directly measured second-order response. That step is deferred (Sec. 3.6); the same caveat touches the magnetic form (Eq. (10)), which is a prescribed geometric plaquette action rather than an effective-action Hessian.

3.3 The constitutive closure

The two geometric forms are not independent. Written through the hop-vector matrix $M = [\mathbf{V}_1 \ \mathbf{V}_2 \ \mathbf{V}_3]$, the electric form is the Gram matrix $\boldsymbol{\varepsilon} = \sum_a \mathbf{V}_a \mathbf{V}_a^\top = M M^\top$, and the magnetic form is the Gram matrix of the pairwise cross products, which is exactly its adjugate.

Theorem 3.2 (Adjugate relation). *For any three spatial hop vectors, the magnetic form is the adjugate of the electric one, $Q_B = \text{adj}(\boldsymbol{\varepsilon})$:*

$$\mu^{-1} = Q_B = \text{adj}(P) = \text{adj}(\boldsymbol{\varepsilon}) \quad (13)$$

Proof. The cross products $\mathbf{V}_a \times \mathbf{V}_b$ are the rows of $\text{adj}(M)$, so $Q_B = \text{adj}(M)^\top \text{adj}(M) = \text{adj}(M^\top) \text{adj}(M) = \text{adj}(MM^\top) = \text{adj}(\boldsymbol{\varepsilon})$. Verified symbolically for general vector components in `src/utilities/electric_induced_action.py`. $\square$

The physically relevant statement is the *proportional* closure $\boldsymbol{\mu}^{-1} = \gamma \text{adj}(\boldsymbol{\varepsilon})$ for a positive scalar γ : the electric and magnetic sectors may carry different overall normalizations (Sec. 3.6), and the exact equality $\gamma = 1$ is the special case realized by the raw lattice forms. Either way, the adjugate carries each eigenvalue of $\boldsymbol{\varepsilon}$ to the product of the other two, so every principal direction obeys $\varepsilon_i (\mu^{-1})_i = \gamma \det \boldsymbol{\varepsilon}$: the impedance product is *isotropic* even though $\boldsymbol{\varepsilon}$ and $\boldsymbol{\mu}^{-1}$ are each anisotropic, in opposite senses about $(1, 1, -1)$. For the lattice values, $\varepsilon_a \mu_a^{-1} = 1 \cdot 16 = 16 = 4 \cdot 4 = \varepsilon_p \mu_p^{-1}$.

3.4 The birefringence verdict

Isotropic impedance is exactly the condition that removes vacuum birefringence. For a plane wave of unit direction $\hat{\mathbf{k}}$ in a medium $(\boldsymbol{\varepsilon}, \boldsymbol{\mu}^{-1})$, Maxwell's equations give $\det(K \boldsymbol{\mu}^{-1} K + \omega^2 \boldsymbol{\varepsilon}) = 0$ with $K = [\hat{\mathbf{k}}]_\times$. With $\boldsymbol{\mu}^{-1} = \gamma \text{adj}(\boldsymbol{\varepsilon})$ this determinant factorizes—for a *general* symmetric $\boldsymbol{\varepsilon}$ and general direction, requiring no shared eigenvectors—as

$$\det(K \boldsymbol{\mu}^{-1} K + \omega^2 \boldsymbol{\varepsilon}) = \det(\boldsymbol{\varepsilon}) \omega^2 (\omega^2 - k^\top \boldsymbol{\varepsilon} k)^2 \quad (\det \boldsymbol{\varepsilon} = 16 \text{ for the lattice blocks}) \quad (14)$$

(the general prefactor is $\det \boldsymbol{\varepsilon}$; the value 16 is that determinant for the lattice blocks, not a universal constant). The $\omega^2 = 0$ root is the longitudinal mode; the physical root is the *doubled* transverse root, so both polarizations share the single dispersion

$$\omega^2 = k^\top \boldsymbol{\varepsilon} k = 3kx^2 - 2kxky + 2kxkz + 3ky^2 + 2kykz + 3kz^2 \quad (15)$$

and the polarization split vanishes identically,

$$\Delta(\omega^2) \propto |\varepsilon_a \mu_a^{-1} - \varepsilon_p \mu_p^{-1}| = |\det \boldsymbol{\varepsilon} - \det \boldsymbol{\varepsilon}| = 0 \quad (16)$$

Gauge-sector vacuum birefringence *cancels*. Because the split depends only on the anisotropy ratios—fixed by the geometry through Theorem 3.2—and not on the overall block scales, the cancellation is invariant under independent rescaling of $\boldsymbol{\varepsilon}$ and $\boldsymbol{\mu}^{-1}$: it survives the proportional closure for any γ , and holds for any hop vectors. This is the one gauge-sector result that is unconditional given the blocks. An engine-level and a purely numerical check corroborate it: `exp_03_geometric_blocks` extracts both blocks from the engine, confirms $\boldsymbol{\mu}^{-1} = \text{adj}(\boldsymbol{\varepsilon})$ from them to 5×10^{-15} , and the Fresnel solver (`exp_03_dispersion_core`) re-

turns $\max |v_+ - v_-| = 1.6 \times 10^{-15}$ over 2000 directions, while a deliberately detuned control (breaking the adjugate relation) returns a split of 0.5. This is an independent code-path verification of the theorem—a separate implementation, but within the same framework and engine—not an independent physical measurement.

3.5 The common-mode speed anisotropy

The null split does not make the sector isotropic. The shared dispersion $\omega^2 = \mathbf{k}^\top \boldsymbol{\varepsilon} \mathbf{k}$ is direction-dependent: the common phase speed $v^2(\hat{\mathbf{k}}) = \hat{\mathbf{k}}^\top \boldsymbol{\varepsilon} \hat{\mathbf{k}}$ ranges over the eigenvalues of $\boldsymbol{\varepsilon}$, $v^2 \in [1, 4]$ —a factor- ~ 2 anisotropy of the vacuum speed about $(1, 1, -1)$, shared by both polarizations. This is a *common-mode* speed anisotropy, not birefringence, and it is not small; the engine blocks give $v \in [1.00, 2.00]$.

Remark 3.3 (Single domain versus O_h restoration). The anisotropy is the response of a single trigonal domain. Averaging the two blocks over the four ways of choosing three of the four cube body-diagonals as hop directions restores full cubic symmetry,

$$\langle \boldsymbol{\varepsilon} \rangle = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \quad \langle \boldsymbol{\mu}^{-1} \rangle = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix} \quad (17)$$

so the O_h -restored medium is isotropic. Note, however, that the *exact* adjugate identity does *not* survive this average: $\text{adj}(3\mathbb{I}) = 9\mathbb{I} \neq 8\mathbb{I}$, so $\langle \boldsymbol{\mu}^{-1} \rangle = \frac{8}{9} \text{adj}(\langle \boldsymbol{\varepsilon} \rangle)$ —it is the *proportional* closure ($\gamma = \frac{8}{9}$) that survives. The averaged medium is non-birefringent because it is *isotropic*, not because the exact adjugate relation persists. The physical lattice, though, has one fixed hop set—its optical axis is a specific body-diagonal—so the four-domain average is a fictitious ensemble it does not realize, and there is no dynamical mechanism that performs it. Whether the common-mode anisotropy is therefore a real prediction (single domain, factor ~ 2) or washed out (restored, isotropic) is undecided; it is a question about vacuum structure, distinct from the polarization verdict, which cancels in *every* domain regardless.

The optical axis itself is geometric: as *axes* the three hop directions are three of the four cube body-diagonals, and $(1, 1, -1)$ is the fourth—the one the hop set omits. Selecting three of four diagonals breaks O_h to the trigonal D_{3d} about the omitted diagonal, and a D_{3d} -invariant quadratic form is uniaxial, which is why both blocks come out $\{\text{axis}; 4, 4\}$.

3.6 Scope: structure, normalization, and the open dynamical tensor

Three items are deferred rather than concealed. First, *normalization*: a single common factor (a one-loop $1/g^2$) multiplying both parts of the action cancels from the source-free equations of motion and so does *not* set the photon speed; the speed is set by the *relative* electric/magnetic normalization—the tick-weight and the temporal-versus-spatial lattice scales—which is the scalar γ of the proportional closure and is left undetermined. Because the birefringence verdict is rescaling-invariant, this does not weaken it, but it means the sector fixes anisotropic *structure*, not the absolute vacuum speed.

Second, the *dynamical* induced action. The blocks above are bare geometric plaquette forms; a genuinely matter-integrated effective action is a loop-level object, and demonstrating that integrating out the tokens reproduces Eqs. (10)–(12)—with these weights and no cross-plaquette terms—is the content of Remark 3.1. The route is the second-order response of the physical band of the one-period transfer operator $T(\mathbf{k})$ summed over the Brillouin zone ($\text{Tr} \ln T$). We do not carry it out and do not claim it. A naive band sum is Ward-divergent along the optical axis (the physical band is exactly flat there, so the intraband denominator $\lambda(\mathbf{k}) - \lambda(\mathbf{k} \pm \mathbf{q})$ vanishes); once the occupation-cancelled diamagnetic term is subtracted, the naive current–current object returns the *wrong* tensor (uniaxial about $(1, -1, 0)$), and the correct assembly is the orbital-susceptibility (Berry-curvature plus quantum-metric) combination. This single-object dynamical extraction is a stated open item, deferred to follow-on work. Since the geometric and effective-action routes must compute the same induced action, and the geometric route already gives Eqs. (10)–(12), finishing it is a matter of methodological completeness, not of deciding the verdict.

The status of the three claims is therefore distinct (Table 1.1): the null polarization split, *conditional* on the blocks, is settled; the derivation of the blocks as the actual dynamical effective action is partial; and the common-mode isotropy is open. Naming the word once more, for precision: the sector predicts the *absence* of gauge-sector vacuum birefringence about $(1, 1, -1)$, together with a common-mode speed anisotropy whose reality turns on the vacuum-structure question above.

4 Conclusion

[TODO: Summarise what was established, what is partial, and what the natural next paper / experiment would be. Avoid rehashing the introduction — the conclusion is for what changed by the end.]

Acknowledgements

[TODO: Acknowledge collaborators, reviewers, funding, and tools. Keep it brief; one paragraph is typical.]

A Code and Data

[TODO: Describe what is in the companion repository: directory layout, how to inspect the data files, where the figure-generating scripts live. The aim is that a reader can reproduce or audit any claim in the paper from this appendix plus the linked DOI.]

B Reproducibility Quickstart

[TODO: Compact reproducibility quickstart. Goal: a reader who has just cloned the repository at the release tag can reproduce the paper’s headline claim in under five minutes.

Suggested structure:

- **Environment.** Operating system, language version, key dependencies (point to the manifest file rather than re-listing).
- **Build the paper.** The commands needed from a fresh clone.
- **Verify the headline result.** A minimal snippet — ideally under ten lines — that re-derives the central numerical claim.
- **Where to look next.** Pointers to the experiment scripts and data files for deeper checks.

]

References

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